



Murtuza Husain

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Gender: Male **Date of birth**: 01/10/2000 **Nationality**: Indian

ABOUT ME

Energetic and results-driven professional with 2.5 years of professional experience, including 1.3+ years as a Software Engineer at Avaya. Currently pursuing an M.S. in Computer Science at the University of Bonn and working as a student assistant at ZB Med. Adept at fostering innovation through collaborative teamwork and delivering high-quality software solutions in dynamic environments. With a strong foundation in both academic research and practical application, I bring a versatile technical skill set and a passion for solving complex challenges. Eager to contribute expertise to impactful and forward-thinking projects in Computer Science.

WORK EXPERIENCE

ZB MED - Informationszentrum Lebenswissenschaften

City: Bonn | Country: Germany

[06/01/2025 – Current] **Student Assistant Computer Science**

Evaluating **rule-based**, **prompt-based**, and **fine-tuned NER models** on agricultural datasets to identify the most effective method for domain-specific entity extraction. Leveraging tools like **spaCy**, **Hugging Face**, and **LLM APIs** to benchmark performance.

Tech Stack: Python, spaCy, Hugging Face Transformers, LLM APIs, Scikit-learn, Pandas, NLTK

GESIS – Leibniz Institute for the Social Sciences

City: Cologne | Country: Germany

[01/03/2024 – 31/12/2024] **Student Assistant Computer Science**

Developed machine learning models for **named entity recognition (NER)** and **relation extraction** in computer science research papers, focusing on ML-related terms.

- Extracted entities such as **ML model mentions**, **dataset names**, **reference links**, and their interrelations.
- Built scripts for **coreference resolution** on annotated datasets to improve entity linking and context tracking.
- Applied NLP techniques and machine learning pipelines to enhance accuracy and contextual understanding.

Tech Stack: Python, spaCy, Scikit-learn, Transformers, Coreference Resolution, NLP, NER, Relation Extraction

Avaya

City: Pune | Country: India

[11/07/2022 – 29/09/2023] **Java Developer**

Designed and delivered a scalable, cloud-native software ecosystem, emphasizing microservices architecture, real-time data processing, and secure REST API integrations.

Key Contributions:

- **API Design & Documentation:**
 - Developed RESTful APIs using **OpenAPI 3.0**, enabling seamless third-party integration.
 - Authored detailed **API documentation** and **use case diagrams** to support internal and external developers.
- **Microservices Architecture:**
 - Built robust **Java Spring Boot microservices** backed by **PostgreSQL**, ensuring modular scalability and performance.
 - Implemented efficient **Spring Data JPA** repositories to optimize database operations.
- **Kafka Data Streaming:**

- Designed **Apache AVRO (AVSC) schemas** for Kafka-based data flow, improving throughput and schema validation.
- **Feature Implementation:**
 - Engineered a **Bulk Agent Management** module supporting Excel uploads via **Apache POI**, enhancing user productivity.
- **Testing & Issue Resolution:**
 - Ensured code reliability through **JUnit-based unit testing** across APIs and services.
 - Proactively resolved **UAT and production issues**, prioritizing based on impact and severity.
- **Cloud Deployment & CI/CD:**
 - Deployed microservices on **Kubernetes** using **Azure CLI** and **Helm**, ensuring scalable and resilient deployment.
 - Used **Liquibase** for database version control and schema migrations.
- **Security & Access Control:**
 - Integrated **Keycloak** for centralized authentication and role-based authorization.

Technologies & Tools:

Java, Spring Boot, REST APIs, OpenAPI 3.0, Kafka, Apache AVRO, Apache POI, PostgreSQL, Spring Data JPA, Liquibase, Kubernetes, Azure CLI, Helm, JUnit, Jenkins, Jira, Keycloak, Microservices

EDUCATION AND TRAINING

[01/10/2023 – 01/10/2025]

MSC Computer Science

University of Bonn

City: Bonn | Country: Germany |

[22/08/2018 – 07/07/2022]

Bachelor of Engineering - Information Technology (AI and ML Honors) CGPA : 9.74/10

Pune Institute of Computer Technology

City: Pune | Country: India |

[30/12/2022 – Current]

Microsoft Certified Azure Fundamentals

Microsoft <https://www.credly.com/badges/c05f1211-d88d-4529-8bf1-ff76e728372c>

[01/09/2020 – 24/12/2020]

DeepLearning.AI Deep Learning Specialization

deeplearning.ai <https://www.coursera.org/account/accomplishments/specialization/2PW8FYWC2YNP>

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

English

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

German

LISTENING A1 READING A1 WRITING A1

SPOKEN PRODUCTION A1 SPOKEN INTERACTION A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Technical Skills

Java | Python | JavaScript | C++ | C | Machine Learning | MySQL | CSS | Hibernate | H TML | Data Structures | Microservices | PostgreSQL | JPA | Apache Kafka | Unit Testing | JSON | REST | Spring Boot | Microsoft Azure | Developing web applications and web technologies | Maven | Software Development | Natural Language Processing | Source control: Git | Identifying and debugging issues

Soft Skills

Adaptability | Empathy | Collaboration | Teamwork | Good Communication | Attention to detail | Critical thinking | Problem-solving | Time management | Patience

PROJECTS

Tourist Spot Recommendation System (Powered by ML and Flask)

In my culminating engineering project, I conceptualized and built a dynamic Full Stack Web Application with a distinctive goal – assisting users in discovering optimal tourist spots. The project's innovation centered around the incorporation of advanced Machine Learning algorithms, particularly a hybrid recommendation system. By seamlessly integrating collaborative filtering and content-based filtering, the application delivered personalized and well-informed suggestions to users. This undertaking not only highlighted my technical proficiency in Full Stack development but also underscored my dedication to harnessing cutting-edge technologies for pragmatic, user-centric solutions.

Medicare(Android Application)

Designed and developed an Android app simplifying doctor location based on user preferences. The app enables users to find doctors or hospitals by region and specialization offline, enhancing convenience and showcasing my commitment to accessible, user-friendly applications.

Pune Metro Booking System(Web Application)

Developed a user-friendly Full Stack Web App as a key part of a DBMS project. Powered by ReactJS, MySQL, and NodeJS, it enables seamless booking, cancellation, and viewing of Pune Metro tickets.

Weather App using Vanilla JavaScript, HTML, CSS

Engineered an intuitive Weather App with a user-centric design using JavaScript. The project marked a significant milestone as I delved into API integration, mastering the art of retrieving real-time weather information and presenting it seamlessly to users

HONOURS AND AWARDS

5th Rank in S.E Bachelor of Engineering University Exam Awarding institution: Savitribai Phule Pune University